Block Sparse Attention

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1 Setup

We evaluate three configurations on the standard suite and report the average accuracy across five random seeds. The method achieves a substantial improvement over the strongest published dense baseline.

$$b = \frac{1}{2}\sum_{i=1}^{n} x_i^2, \qquad A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}. \tag{1}$$

Sparse attention reduces the quadratic cost of the attention operator to a linear one under mild assumptions on the block structure of the mask. Information retrieval benefits from the same sparsity pattern at every depth of the network.

The kernel is implemented once and reused for both training and inference. Training with the sparse kernel from scratch converges in the same number of steps as the dense baseline.

2 Method

Each block is processed independently before the aggregation step that combines the partial attention scores. We report the mean and the standard deviation over five random seeds for every configuration in the study. The routing scheme assigns every query block to a small set of key blocks chosen by a learned scorer.

$$\lambda = \frac{1}{n}\sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2. \tag{2}$$

Ablations show that removing the scorer costs four points of accuracy on the longest split. The kernel is portable and needs no vendor specific intrinsics to reach the reported speed.